

current flows when there are no apertures and Fig. 7 shows how the current flow gets disturbed when there is an aperture.

We need to remember that most electronic systems need ventilation slots for heat dissipation, which can be a problem for the EMC/EMI. One of the solutions is to provide the ventilation through a cluster of holes, which allows the induced current flow undisturbed as shown in Fig. 8.

Another challenge in enclosures are the seams. Seams are inevitable in sheet-metal enclosures as the enclosures are made of multiple sections. When the sections are fastened by rivets or screws they make the seams leaky due to apertures. If these seams have to be made leak-proof three things need to be done:

1. The adjoining sections should overlap.
2. The rivets and screws should be as close as possible to reduce the aperture effect. This is where welding helps!
3. The two overlapping surfaces have to be conductive and, for better shielding, EMI gaskets or EMI fingers should be used.

A challenge that designers face in electronic packaging very often is that a solution to one problem leads to another problem (EMI/EMC vs heat dissipation). However, sheet-metal enclosures lend themselves to EMC/EMI problem in a natural way, while plastic enclosures pose a big challenge for EMC/EMI.

To make plastic enclosures meet EMC/EMI needs these have to be made conductive. Most popular but expensive solution is to coat a conductive paint inside the plastic enclosure, or to make the plastic conductive with additives. Other methods are electroless plating or metal-foil lining, both of which are expensive and delicate.

Another approach is to enclose the PCB assembly in a metal shield so that the plastic enclosure does not have to be treated for EMC/EMI. This approach is popular in radio products as sometimes power levels are high and need to be contained in the PCBA module itself. PCB-mounted shielding can be custom designed, or ready-to-use shields can be tried. Most Wi-Fi

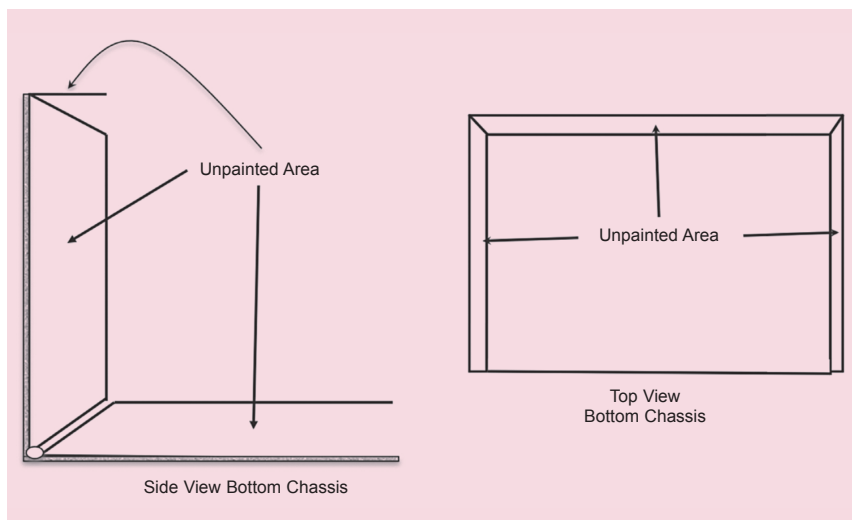


Fig. 9: Cross-sections of an enclosure

and Bluetooth modules come with shields to reduce extraneous radiation and their PCBA is shielded, except the antenna.

The best way to make a product EMC/EMI-standards compliant is by following the six-step process mentioned below:

1. Understand the standards compliance needs with respect to radiation and budget 3dB more for safety margin. Just meeting the limit is perfectly fine with certification process but due to manufacturing process there may be variations, which can fail the product.

2. Ensure the enclosure is well designed, keeping the above-mentioned information in mind. But if off-the-shelf enclosure is being used, ensure right enclosure is chosen. More than the selection, follow the process recommended by the vendor for assembly. *(I remember we had used an on-PCB shield as EMI solution and the product passed the limits with good margin. However, during production we found the products failing sample tests. We walked through the production line and found that the stage where on-PCB shield was being installed had an issue. Due to margin error in the holes in PCB the shield could not be inserted easily, and the technician was pushing the shield in using a nylon mallet. It was a perfect way to do, but the mild nylon mallet blow opened up the shield's seam on the sides and lead to radiation failure! In case of FCC certification, when you*

*get an FCC registration number for compliance, it intrinsically means that FCC can take samples from the market and test your product for radiation compliance. So, EMC/EMI certification is a fairly intense job.)*

3. Electronics circuit design and PCB layout as per design guidelines is important but fairly intense and cannot be covered here.

4. First version of the product should be subjected to a pre-scan in a lab to get the measurements. This will give a telltale story of the product in its current form. Solution can be implemented without much time and effort in the next revision.

5. When the scan (both pre and final) is done by rotating the product 360° around vertical Y axis, the reading shows up any seam- or aperture-related leakages. Some labs do it in a static way with product rotated 90 degrees on its four sides. This can sometimes mask the radiation issue, so it is better to do the scan with a rotating table.

6. From the reading the source of radiation leakage can be identified. A normal spectrum analyser reading will have frequency on the X axis and radiation level on the Y axis; angle has to be manually observed by the compliance engineer and design engineer. Designs based on sheet-metal enclosures can fail due to fabrication and design mistakes like gaps in corner bends from where radiation can leak. The best solution is to have an overlap between top and bottom of the sheet-